

Prof. David L. Azevedo
Instituto de Física
Universidade de Brasília
Campus Darcy Ribeiro, Asa Norte
Brasília, DF, 70910-900
Brazil

April 24, 2026

The Editors
Nature
Springer Nature
The Campus, 4 Crinan Street
London N1 9XW, United Kingdom

Dear Editors,

I am pleased to submit the enclosed manuscript, entitled “**Universal phenomenology, divergent mechanism: a behavioural gradient of mobility from foraging to commute**”, for your consideration as an Article in *Nature*. This work asks a question that the mobility and movement-ecology communities have circled around for two decades but never answered directly: *do the scaling laws that govern human mobility in cities also describe the movement of non-human animals, and if so, what do the differences tell us about the underlying mechanism?*

To answer this, I applied a single estimator battery — covering the four canonical human-mobility laws of González et al. (*Nature*, 2008), Song et al. (*Nat. Phys.*, 2010), Alessandretti et al. (*Nature*, 2020) and Schläpfer et al. (*Nature*, 2021) — to GPS records of 14 African elephants, 25 Northern gannets and 92 white storks tracked for up to a decade, and compared the results with the published human values. I report three findings that, taken together, reshape the current view of mobility as a human-specific phenomenon. **First**, the phenomenological architecture is universal: all three species exhibit truncated power-law step-length distributions, logarithmic saturation of the radius of gyration, four to five hierarchically nested lognormal containers and circadian peaks of return, in precise analogy to the published human patterns. **Second**, the microscopic mechanism is not universal but graded: the Song preferential-return scaling relations hold tightly only as the visitation Zipf exponent ζ approaches unity, defining a behavioural gradient from the home-ranging elephant ($\zeta = 0.28$) through colonial seabirds ($\zeta \approx 0.9$) to the commuting human ($\zeta = 1.0$), and the $\eta = 2$ visitation law is a regime rather than a universal — animals at near-constant locomotor speed return $\eta \approx 1$, whereas humans on motorised infrastructure return $\eta = 2$. **Third**, a falsifiable prediction bears out: when the stork record is restricted to its obligate central-place breeding window, its step-length exponent ($\alpha = 1.70 \pm 0.05$) becomes statistically indistinguishable from the human value (1.75 ± 0.15), demonstrating that a single bird species can cross from the animal regime into the urban-human regime as its central-place obligation varies.

I believe this result belongs in *Nature* because it reframes three separate literatures — urban scaling, Lévy-flight foraging and optimal foraging theory — as limits of a single behavioural-energetic axis. It reconciles Christaller’s central-place theory with Charnov’s marginal-value theorem under one physical principle (conservation of effort per unit time, at regime-dependent velocity variance), offers a testable prediction for how emerging urban-planning paradigms such as the “15-minute city” should reshape the human visitation exponent η towards the animal regime, and supplies a reinterpretation of the long-running Lévy-flight controversy that is consistent with

both the proponent and the sceptic positions. The methods are publicly shareable and all data are already CC0-licensed on the Movebank Data Repository, so the analysis is fully reproducible; the preregistration, code and processed outputs are deposited alongside the submission. I expect the manuscript to interest readers in statistical physics, movement ecology, urban science and behavioural biology.

This work is original, has not been published elsewhere and is not under consideration at any other journal. I have no competing financial or non-financial interests to declare. Companion manuscripts describing methodological extensions are in preparation but not yet submitted; I would be happy to disclose them on request. In accordance with *Nature's* editorial policy, any referee suggestions and exclusion requests have been entered in the designated fields of the online submission system; the selection of referees and of the handling editor rests entirely with the editorial team.

I thank you for considering this manuscript and look forward to your response.

Yours sincerely,

Prof. David L. Azevedo
Instituto de Física
Universidade de Brasília
david888azv@unb.br